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15

AXIALLY SYMMETRIC FLOWS

15.1 Procedures for axially symmetric flows

An axially symmetric flow is similar to the two-dimensional flow in that it has two independent spatial variables and the discussion of grids in §14.1 applies also to these flows. Program DSMC2A is a version of DSMC2 for application to axially symmetric flows. Instead of modelling unit width of the two-dimensional flow, it models the full azimuth of 2π about the axis which is chosen to lie along the x-axis. The y-coordinate becomes the radius and there are obvious consequences with regard to the cell volumes and surface areas. However, it is no longer sufficient to simply ignore the motion in the z-direction and the three-dimensional character of the molecular trajectories must be taken into account.

Advantage is taken of the symmetry to transform, at each time step, the initial point of each trajectory element to the zero azimuth plane. These transformations are similar to those for the one-dimensional flows with cylindrical or spherical symmetry. Consider a molecule at radius y_1 with velocity components in the radial and circumferential directions of v_1 and w_1, respectively. If the molecule moves for time Δt, the new radius is

$$y = \{(y_1 + v_1\Delta t)^2 + (w_1\Delta t)^2\}^{1/2}. \tag{15.1}$$

Also, the velocity components must be rotated so that v remains the radial component. The new values of the radial and circumferential components are

$$v = \{v_1(y_1 + v_1\Delta t) + w_1^2\Delta t\}/y$$

and

$$w = \{w_1(y_1 + v_1\Delta t) - v_1 w_1\Delta t\}/y. \tag{15.2}$$

These equations are implemented in the subroutine AIFR. Program DSMC2A is restricted to surfaces that lie along the cell boundaries, so that they are either parallel to or normal to the axis. Molecular collisions with a normal surface are due to the axial velocity component and the two-dimensional logic remains valid. The parallel surfaces are cylindrical and the three-dimensional logic that led to eqn (12.12) applies also to this flow. This leads to complex code and more general surfaces would involve intersections with conical elements for which the parameters could change at each cell. The logic then becomes even more complex and there is an increase in the computation time. Cell systems such as those used in the G2 codes require this logic for all interactions with cell boundaries and the computation time is prohibitive if the three-dimensional logic is employed.

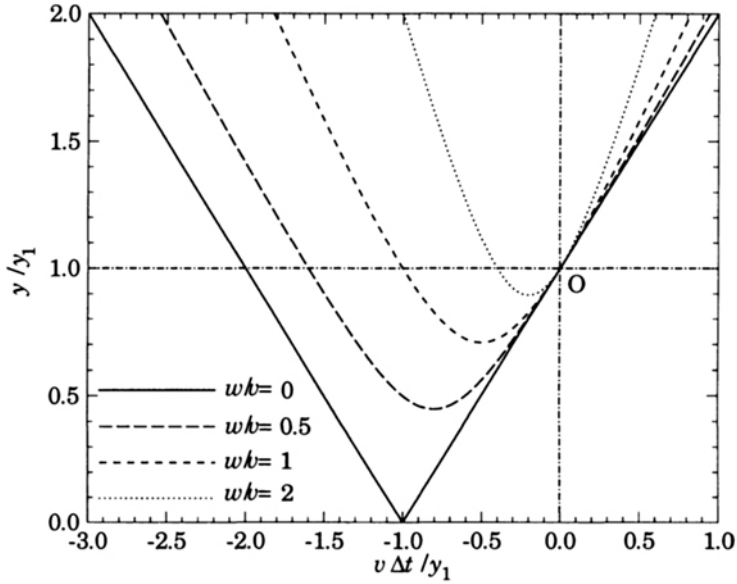


Fig. 15.1 The projection of typical trajectories on the normal plane.

The alternative to the exact three-dimensional logic is to subdivide the basic time step into a sufficiently large number of intervals that two-dimensional logic can be applied at each of the resulting sub-steps. Note that this is in regard to possible intersections with surfaces during the sub-step and that the transformations of eqns (15.1) and (15.2) are still applied at the end of each sub-step. The problem is to determine the necessary degree of subdivision.

Eqn (15.1) may be rearranged to give the following expression for the ratio of the radius y after time Δt to the initial radius y_1 as

$$y/y_1 = [1 + 2vt/y_1 + \{1 + (w/v)^2\}(vt/y_1)^2]^{1/2}. \tag{15.3}$$

This result is illustrated in Fig. 15.1. For negative values of v, the radius ratio has a minimum value of

$$y_m/y_1 = |w/v|/\{1 + (w/v)^2\}^{1/2}$$

after a time interval of

$$v\Delta t/y_1 = -\{1 + (w/v)^2\}^{-1}.$$

It seems reasonable to restrict the size of the time sub-step to one fourth the time interval to this minimum with some restriction on the minimum sub-step when v is very small. Also, for both positive and negative values of v, the time sub-step may be restricted to one sixth the value of $y_1/|w|$. This set of criteria has been employed in the axially symmetric version of the G2 program system and has been found to give satisfactory results.

15.2 Radial weighting factors

The most severe practical problem associated with DSMC calculations for axially symmetric flows is the small fraction of the molecules that are located near to the axis. For example, if there are 50 equally spaced cells in the radial direction in uniform flow, the sample size in the outermost cell will be 100 times greater than that in the cell at the axis. This has led to the introduction of *radial weighting factors* such that a molecule located far from the axis represents more real molecules than one near the axis. This can even up the number of molecules per cell but, since the weighting factor will change along the molecular trajectories, there is the probability that each molecule will have to be either discarded or duplicated at every time step. Weighting factors will obviously lead to fewer problems if the flow is primarily directed away from the axis and the predominant effect of the factors is that molecules are discarded rather than duplicated.

The weighting factors may be based either on the radius of the cell in which the molecule lies or on the radius of the molecule itself. The former case leads to errors when a molecule that is moving parallel to the axis moves to a cell with a different height and therefore a different weighting factor. In addition, this procedure has been found to lead to general difficulties in maintaining smooth flow gradients normal to the axis. Cell-based factors are not recommended. The alternative molecule-based factors would appear to require inefficient and complex collision coding in that the model molecules in the cell represent different numbers of real molecules. However, it has been found that there is a negligible effect on the flow if all the molecules in a cell are regarded as having an average weighting factor in the collision routines. The average weighting factor then multiplies the number of molecular choices, but each collision counts as a number of identical collisions equal to this factor, and the collision procedures need not be changed.

Program DSMC2A employs weighting factors that are equal to the ratio of the molecule radius to a reference radius RWF that is set in the data. This reference radius may be set to a small fraction of the height of the inner cell and all cells in the above hypothetical case would then contain the same number of simulated molecules. For very small values of RWF, a molecule that moves very close to the axis would be multiply duplicated and the logic in the subroutine WEIGHT allows for this. The possibility of having identical molecules in a cell forced several modifications to the subroutine SELECT that has been common to almost all the programs. The undesirable impact of the molecule duplication may be lessened in a steady flow by the imposition of a time delay on the appearance of the duplicate molecule. This has been implemented in this program through the 'molecule duplication buffer' which contains MNB molecules. Each duplicated molecule is first generated at a random address in this buffer. The molecule that was previously at this address is then placed into the flow, but this is after a random delay with an average value that depends on the size of the buffer.

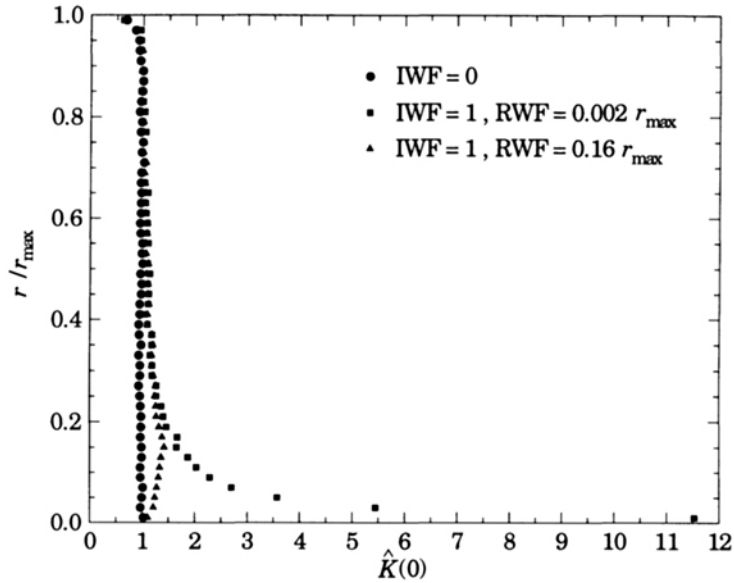


Fig. 15.2 The normalized mean square density fluctuation.

Radial weighting factors have been widely used in applications of the DSMC method to axially symmetric flows. However, the results have frequently exhibited a greater degree of scatter than would be expected from the usual criteria based on sample size or from experience with two-dimensional calculations with similar geometry. Program DSMC2A has therefore been used to determine typical normalized mean square density fluctuations in a uniform stationary gas with and without weighting factors and for various values of the reference radius RWF.

The test cases employ 50 cells of uniform height in the radial direction. The axial length of the flowfield is equal to the radius, but the mean square fluctuation has been averaged in the axial direction. Calculations were made without weighting factors, for weighting factors with a reference radius of 0.002, and for weighting factors with a reference radius of 0.16. The results from the three cases are shown in Fig. 15.2. For no weighting factors (IWF = 0), the normalized mean square density fluctuation has the expected value of unity, although there is some reduction in the scatter near the outer stream boundary. The second case employs weighting factors and a reference radius of one tenth the cell height. The number of molecules in the innermost cell then increases by a factor of 50 and, if the normalized mean square fluctuation remained unity, the scatter would be reduced by a factor of about seven. However, Fig. 15.2 shows that the normalized mean square fluctuation in the cell nearest the axis increases by more than a factor of ten. This shows that the use of radial weighting factors can actually lead to an increase in the scatter at the axis.

The third test case with a reference radius of 0.15 improves the sample in the innermost cell by a factor of just over three. There is only a slight increase in the normalized mean square density fluctuation in this cell so that there is a clear gain. There is an increase in the scatter of approximately 50% at the reference radius. However, there is a factor of four increase in the sample at this radius over the no-weighting-factor case and there is a net gain at this radius also.

The use of the very small reference radius in order to obtain similar samples in each cell was not only counterproductive in terms of the scatter, but led to a systematic reduction of the order of 10 to 20% in the density at the axis. The calculation was repeated with an axial velocity corresponding to a Mach number of three and, while the excess scatter was reduced by about 40%, the systematic error in the density was not improved. If radial weighting factors are used, the reference radius should be very large in comparison with the radius of the innermost cell. Even then, there may be a slight anomaly in the flow properties at the reference radius.

15.3 Flow past a flat-nosed cylinder

Click for referenced Color Plate

The listed data in the program DSMC2A is for the flow of argon at a number density of 10^{21} m^{-3} and a temperature of 100 K at a velocity of 1000 ms^{-1} past a flat-nosed cylinder. The radius r_c of the cylinder is 0.01 m so that the overall Knudsen number based on the diameter of the cylinder is 0.0474. The Mach number of the flow is 5.37 and the conditions are similar to those for the two-dimensional vertical plate that was the subject of §14.6. The temperature of the flat face is 1000 K which is close to the continuum stagnation temperature of 1060.5 K and this case will be referred to as the 'adiabatic case'. The temperature of the cylindrical surface is 300 K and the diffuse reflection model is assumed.

The centre of the flat face is located at the origin and the extent of the simulated flowfield in the axial direction is from –0.02 m to 0.02 m, while it extends to a radius of 0.03 m. The cell grid is 80×60 with equal spacing in each direction. The cell width and height is approximately equal to one-half a mean free path and there are four sub-cells per cell. The maximum density is just over four times the freestream density and this occurs at a temperature of approximately ten times the freestream density where the mean cross-section for argon is half that at the stream temperature. This means that the mean free path in the stagnation region is approximately equal to the cell dimension. The grid dimensions are marginal but, given that the sub-cell dimension is then one-half a mean free path, this should not lead to any error. A 'cold surface' calculation was also made with the temperature of the flat surface equal to 300 K. The cell size was excessive for this calculation and flows with large density increases in the stagnation region require a more sophisticated grid. Weighting factors were employed for these calculations with the reference radius RWF set to 0.004. This is eight times the cell height and this choice corresponds to the value that led to satisfactory results in the preceding section.

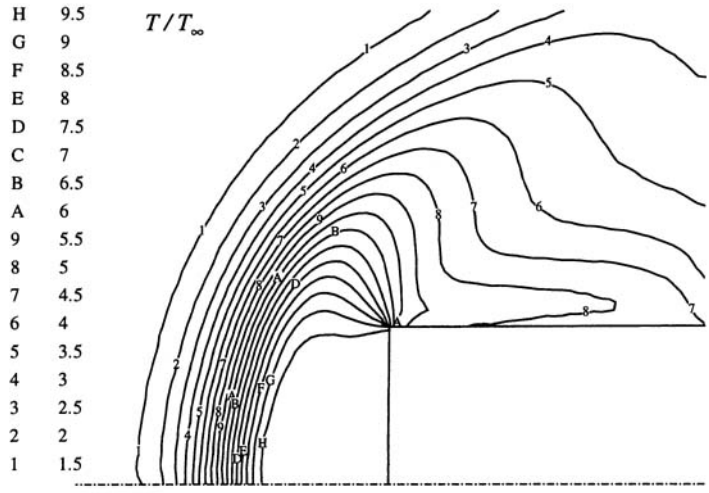


Fig. 15.3 Temperature contours in the adiabatic surface case.

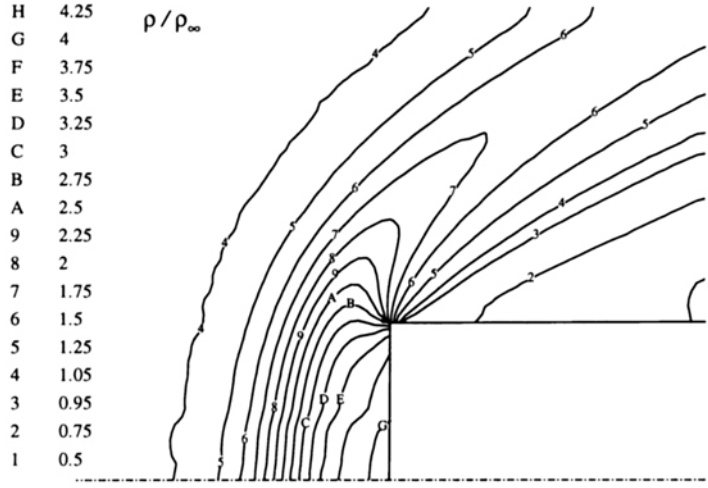


Fig. 15.4 Contours of constant density in the adiabatic surface case.

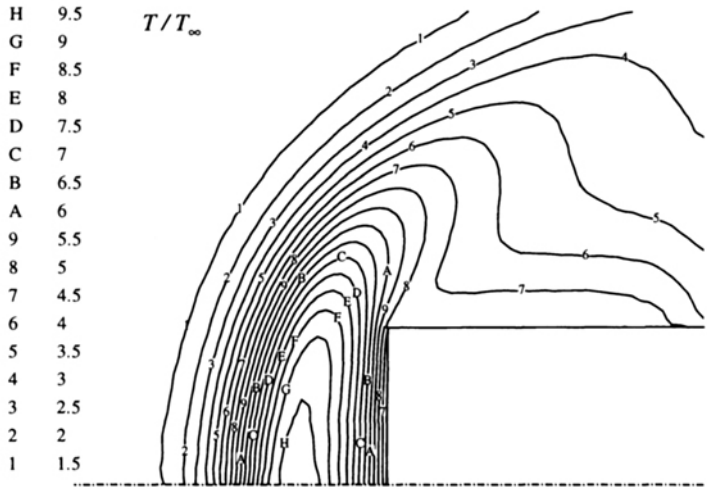


Fig. 15.5 Temperature contours in the cold-surface case.

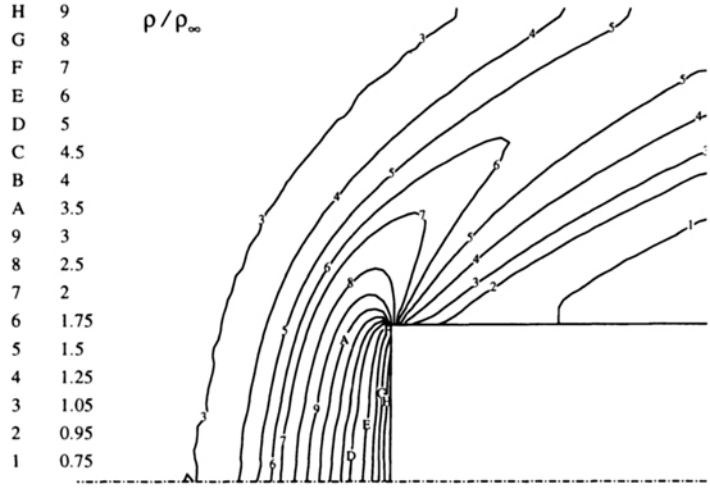


Fig. 15.6 Contours of constant density in the cold-surface case.

The temperature and density contours for the adiabatic case are shown in Figs. 15.3 and 15.4, respectively. These are generally similar to Figs 14.26 and 14.28 for the two-dimensional flow past a vertical flat plate. The shock stand-off distance may be defined by the intersection with the axis of the contour $\rho/\rho\infty = 2.43$ that lies at the centre of the normal shock wave. This occurs in the adiabatic case at 0.34 times the diameter of the cylinder. This may be compared with the figure of 0.78 that is obtained from Fig. 14.28 for the two-dimensional flow.

The corresponding contours for the cold-surface case are shown in Figs 15.5 and 15.6. There is a significant increase in the density and a decrease in the temperature in the thermal boundary layer adjacent to the surface. The shock stand-off distance drops to 0.31 times the diameter of the cylinder. The cooling effect extends around the shoulder of the body, but the flow properties well away from the stagnation region are not greatly affected.

The pressure distribution along the flat surface is shown in Fig. 15.7 through the ratio of the pressure coefficient to the continuum stagnation pressure ratio $(C_p)_0$ as a function of the radius ratio. The pressure near to the axis is slightly above the continuum pressure, but the area represented by this region is small and the pressure is significantly below the continuum stagnation pressure over most of the face. The average pressure is three to four percent below the stagnation pressure. This result is consistent with the observed pressures (Potter and Bailey 1963).

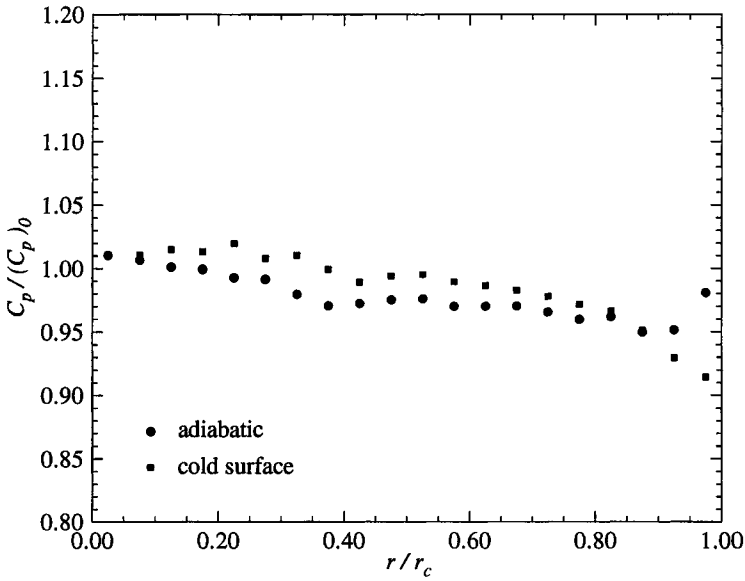


Fig. 15.7 The pressure distribution along the flat face.

15.4 The Taylor–Couette flow

Click for referenced Color Plate

A Couette flow with one stationary surface and one moving parallel surface can be generated by program DSMC2A in five distinct ways. Two coaxial cylinders could have a relative velocity in the axial direction with either the inner or outer cylinder moving. Alternatively, either the inner or outer cylinder could rotate about the axis with a fixed angular velocity and give its surface a circumferential velocity. Finally, one of two plates normal to the axis could rotate about the axis. The Taylor–Couette flow is concerned with the case in which the inner cylinder has a circumferential velocity. Taylor found that, at some critical angular velocity, the flow becomes unstable and develops annular vortices. Stefanov and Cercignani (1993) have shown that the DSMC method can be used to study these vortices and program DSMC2A has been used to re-calculate one of their cases.

This case employs an outer cylinder with a radius r_2 that is double the radius r_1 of the inner cylinder. The gas is a simple monatomic gas of hard sphere molecules. The gas is initially stationary and uniform with a density such the mean free path is $(r_2 - r_1)/50$. At zero time, the inner cylinder instantaneously acquires a constant angular velocity such that its circumferential velocity is equal to three times the most probable molecular speed. The flow is therefore characterized as having a Knudsen number (Kn) of 0.02 and a speed ratio s of 3.0. The Taylor number is defined by

$$(Ta) = \frac{4\,\rho^2\,\omega^2\,r_1^4}{\mu^2\,\{1-(r_1/r_2)^2\}^2}, \qquad (15.4)$$

where ω is the angular velocity of the rotating cylinder. Stefanov and Cercignani quote a Taylor number of 521,600 for this case and this is well above the critical number of 33,110. However, for the 'supersonic' surface speed that is employed here, large density and temperature variations develop within the flow and the effective or local values of the dimensionless parameters may be quite different from their nominal values.

For the purposes of the computation, the flow must have a limited length in the axial direction. This was chosen to be equal to the diameter of the outer cylinder so that, as shown in Fig. 15.8, the computational region has an aspect ratio of four. The cylindrical surfaces are specified as being diffusely reflecting with an accommodation coefficient of unity, while the end-planes are specularly reflecting and can be regarded as planes of symmetry. The cells were of a uniform size with a grid of 200 in the axial direction and 50 in the radial direction, with two sub-cells per cell in each direction. There were a total of 120,000 simulated molecules in the calculation. These numbers are larger than those that have been employed in previous examples and there was a problem in keeping within the 8 Mb limit that has been set for the examples. The program has therefore been modified so that memory is not wasted on the storage of the molecular species in a simple gas or on the rotational energy for a monatomic gas. The changes are quite straightforward, but extra care must be taken in setting the variables in the PARAMETER statements.

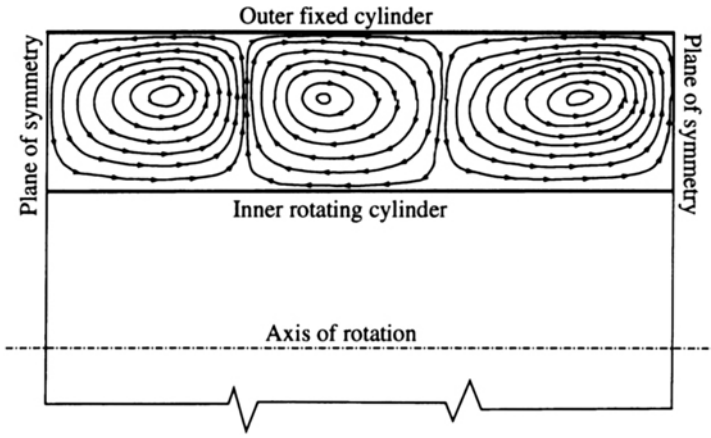


Fig. 15.8 Configuration of Taylor vortices for $s = 3$, $(Kn) = 0.02$.

Fig. 15.8 shows the projection of the streamlines onto the 'normal' plane that contains the axis and is normal to the rotating surface. This is a time average over the 21st to the 30th revolution of the inner cylinder. The apparently stable configuration of three counter-rotating annular vortices evolved through an unsteady process that is illustrated in Fig. 15.9.

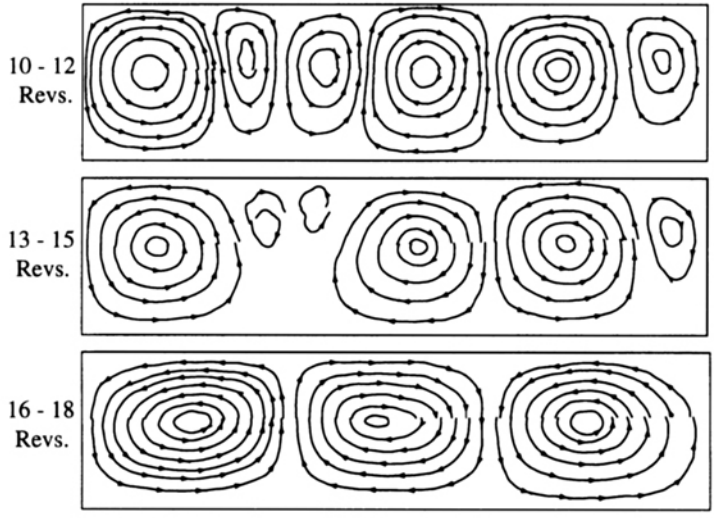


Fig. 15.9 Evolution with time of the Taylor-Couette flow.

The initial state was a uniform gas of density ρ_i in equilibrium with stationary cylinders at the temperature T_w. The rotation of the inner cylinder started impulsively at zero time. A downward moving jet of gas was apparent during the first few revolutions and this developed into the vortex at the left. The other vortices devolved subsequently and, over the time interval corresponding to 10 to 12 revolutions, there were six vortices in three ordered pairs of one large and one small vortex. The three small vortices decayed over the next three revolutions and, after 17 revolutions, there were only three vortices of approximately equal size. The right-hand vortex became larger than the other two by 20 revolutions and it appeared that a single vortex could eventually develop. However, the relative sizes remained constant over the nine revolutions that led to Fig. 15.8 and this indicates that the three-vortex structure could well be stable for this configuration. The results of Stefanov and Cercignani show five vortices for this case, but they did not report the time interval over which the flow was sampled.

As in the one-dimensional high-speed Couette flow of §12.2, the high surface speed of the cylinder leads to a significant heating of the gas. The gas temperature increases until the heat transfer to the surfaces equals the work done in moving the surface. The highest temperatures are near the rotating surface and Fig. 15.10, for the centre vortex, shows that the vortices lead to significant axial gradients in the temperature.

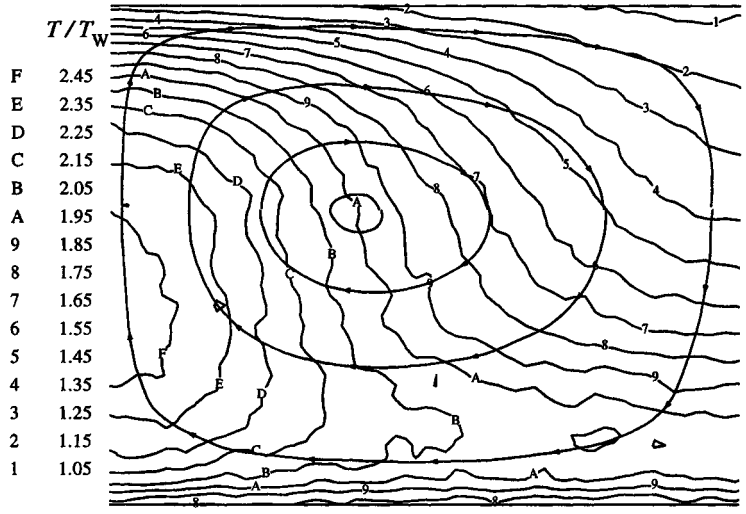


Fig. 15.10 The temperature contours in the centre vortex.

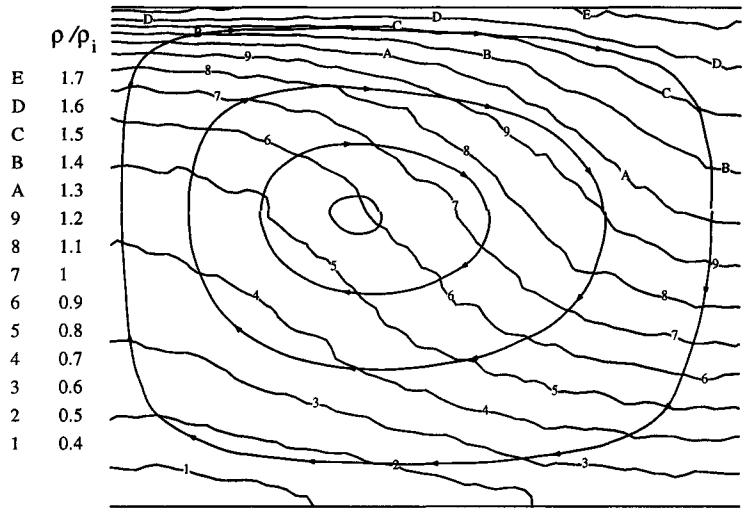


Fig. 15.11 The density contours in the centre vortex.

The density contours of Fig. 15.11 show that the high temperatures correspond to low densities, but the variation in the product of density and temperature shows that there are significant pressure gradients within the flowfield. Sufficient streamlines are shown in these figures to enable the flow features to be related to the vortex. These streamlines are generated as upper and lower segments and the accuracy with which the segments meet indicates that the flow was very nearly steady over the sampling time interval. The results are similar for all three vortices.

The distribution of the circumferential velocity component is shown in Fig. 15.12, and that of the 'vortex' speed in the normal plane is shown in Fig. 15.13. Both of these have been normalized by the most probable molecular speed in the gas at temperature T_w. Note that this is a fixed value and, while these quantities are expressed as speed ratios, they are different from the local speed ratio that takes the temperature variation into account. The speed ratio of the inner cylinder is three and there is significant velocity and temperature slips at this surface. The slips are much less at the fixed upper surface. The speed ratio of the rotational motion in the annular vortices ranges from 0.05 to 0.35. These values are consistent with those reported by Stefanov and Cercignani. The higher speeds in the vortex occur in the lower-density regions of the flow.

While the sampling procedures in the programs are intended primarily for the study of a large time steady flow, the calculation is always unsteady with a physically significant time parameter. Figure 15.9 shows that the programs can be used to obtain information on unsteady processes.

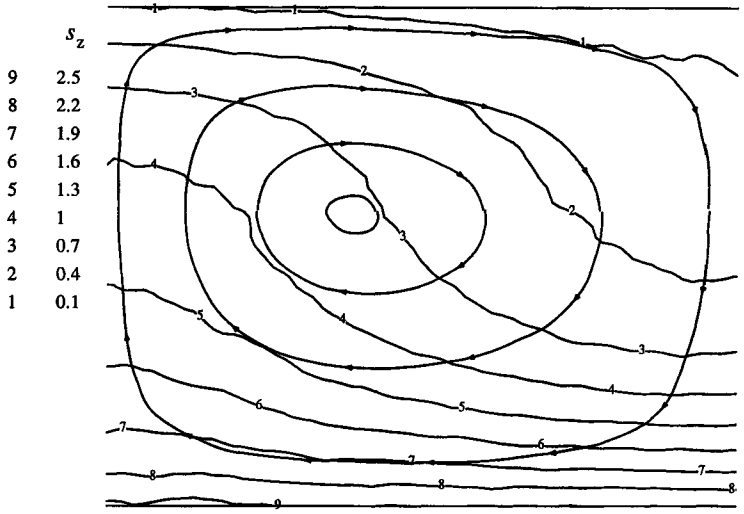


Fig. 15.12 Contours of the circumferential velocity component.

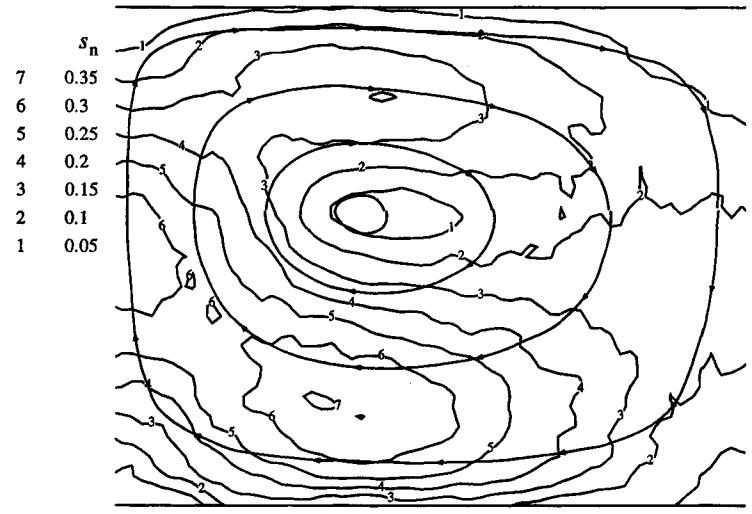


Fig. 15.13 Contours of the rotational speed in the vortex.

The Taylor number that is defined in eqn (15.4) is the square of the Reynolds number

$$(Re) = \frac{\rho u_w \hat{r}}{\mu}. \tag{15.5}$$

Here, $u_w \equiv r_1\omega$ is the surface speed of the rotating cylinder and the characteristic radius is defined by

$$\hat{r} = \frac{2r_1}{1-(r_1/r_2)^2} = \frac{2r_2(1-\Delta r/r_2)}{2\Delta r/r_2-(\Delta r/r_2)^2}, \tag{15.6}$$

where Δr is the distance $r_2 - r_1$ between the cylinders. The value of this Reynolds number for the computed case is 722.

Eqns (1.1), (4.23), (4.62), and (4.65) enable the Reynolds number in a gas of VHS molecules to be written in terms of the speed ratio and Knudsen number as

$$(Re) = \frac{2(5-2\omega)(7-2\omega)}{15\pi^{1/2}} \frac{s}{(Kn)}. \tag{15.7}$$

For the hard sphere gas that has been used in this calculation, eqn (15.7) becomes

$$(Re) = \frac{16}{5\pi^{1/2}} \frac{s}{(Kn)} = \frac{1.8054\, s}{(Kn)}.$$

In the application of this equation to a rarefied gas, there is a case for basing the Knudsen and Reynolds numbers on Δr rather than $\hat{r}$. The values of these numbers for this case are then 0.02 and 271, respectively.

Additional calculations were made to investigate the upper limits of the Knudsen and the lower limits on the speed ratio at which Taylor vortices form. It was found that, at the speed ratio of 3, the Knudsen number could not be increased much above 0.025. However, the width of the flow could be decreased significantly and most calculations were made with just 30 initial mean free paths between the planes of symmetry. This permitted the speed ratio to be increased while still maintaining a reasonable sample near the rotating cylinder. It was found that the single vortex split into two (with the dividing plane normal to the axis) at a speed ratio of about 10. The density was increased and it was found that there was a similar splitting of the vortex at a Knudsen number of approximately 0.003.

Reichelmann and Nanbu (1993) have also applied the DSMC method to the Taylor–Couette flow in air. They employed a rotational speed such that the surface speed ratio was just below 0.5, and calculations were made for Knudsen numbers that ranged from 0.037 to 0.00435. These conditions were chosen to match the experiments of Kuhlthau (1960). The establishment of the Taylor vortices was detected through the behaviour of the torque coefficient on the fixed cylinder. The predictions of the DSMC calculations were in excellent agreement with the measurements.

15.4 Normal impact of a supersonic jet

This application involves the impact of an axially symmetric supersonic jet on a plane normal to the axis. A uniform Mach 5 jet of oxygen with a radius of 0.2 m is directed at a disk of radius 0.6 m that is normal to the axis and is located at 1 m from the exit plane of the jet. The number density of the jet is 10^{20} and its temperature is 300 K, while the disk is diffusely reflecting at a temperature of 1000 K. The mean free path in the jet is 0.01294 m, so that the jet radius is approximately 15.5 mean free paths.

The jet and disk were located at opposite ends of the flowfield which extended to a radius of 0.8 m. The grid was 100 cells in the axial direction and 80 in the radial direction with two sub-cells per cell in each direction. Weighting factors were employed with a reference radius of 0.1 m. The boundaries were specified as vacuum boundaries.

The streamlines of the resulting flow are shown in Fig. 15.14. A normal shock wave forms just upstream of the stagnation point and the sharp 90° deflection of the streamlines occurs in the subsonic region behind this wave. However, the most striking feature of the flow is the very large deflections of the streamlines from near the outer edge of the jet. The Mach number contours in Fig. 15.15 show that these deflections occur in a subsonic region that extends back to near the lip of the jet.

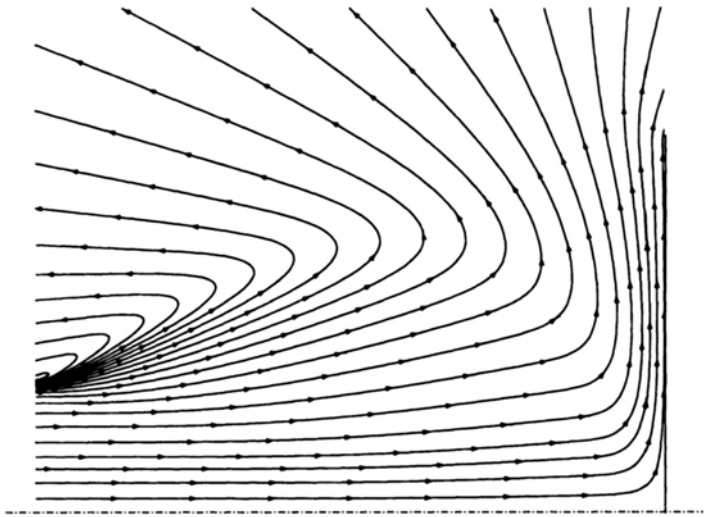


Fig. 15.14 Streamlines in the impact of a $(M\alpha) = 5$ jet on a disk.

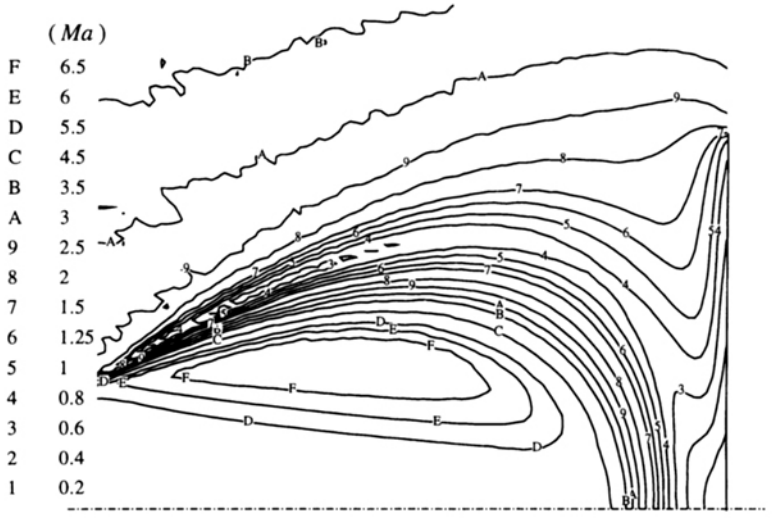


Fig. 15.15 Contours of constant Mach number

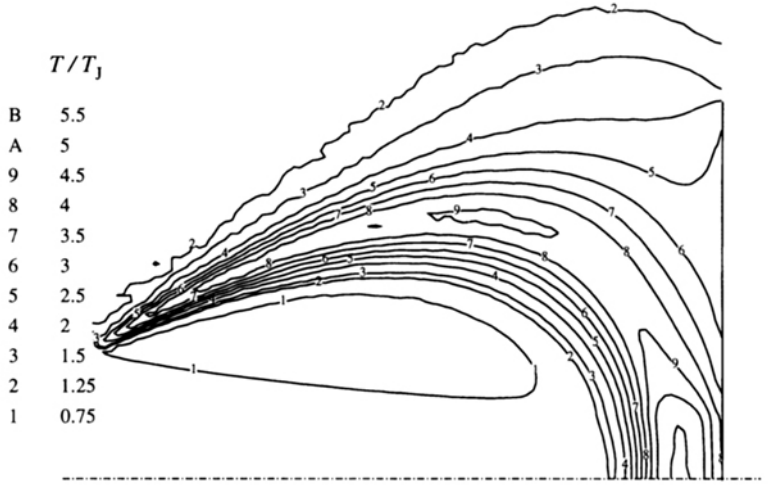


Fig. 15.16 Contours of constant temperature.

The temperature contours are shown in Fig. 15.16 and these are generally similar to the Mach number contour. The subsonic turning region is preceded by an oblique shock wave and is followed by expansion waves. The density contours of Fig. 15.17 are qualitatively different. They are influenced primarily by the expansion from the edge of the jet. There is a large density increase in the stagnation region, but the density increase associated with the oblique shock is relatively small. The subsonic turning region is associated with a 'turning back' of the density contours as the flow subsequently expands supersonically.

The Mach number increases to approximately seven in the outer part of the jet, but the normal shock wave precedes the arrival of the expansion at the axis. The degree of initial expansion of the jet would increase as the disk was moved further from the exit plane.

The extraordinary feature of the subsonic turning region is that it is actually thinner than the shock wave that precedes it. Consider this region at its maximum radius. The density is 0.03 times the initial density ρ_J in the jet and the temperature is about four times the jet temperature. The mean free path is therefore more than ten times larger than the width of the subsonic region. The process is therefore one of *molecular scattering* rather than the quasi-continuum process that is implied by the use of streamlines to describe the flow. The collisions that cause a flow deflection in a highly rarefied gas can occur well away from the site of the deflection.

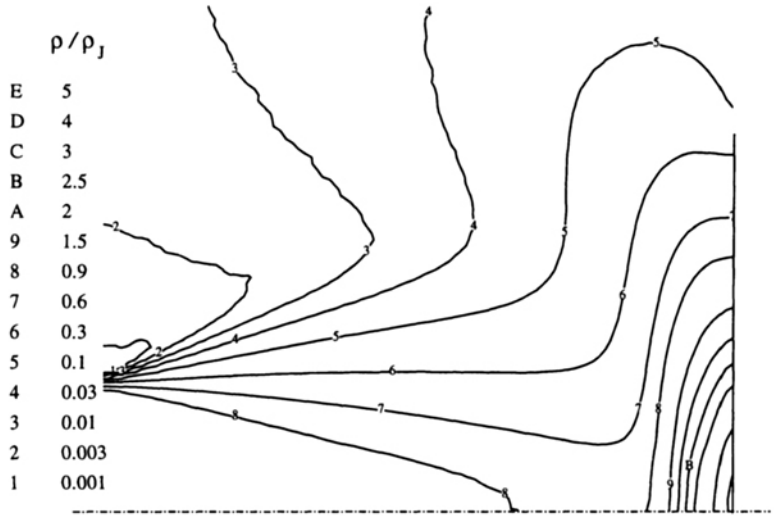


Fig. 15.17 Contours of constant density.

The calculation was repeated for a density three times higher than in the above case. The shock wave thickness decreased proportionally, but the size and shape of the subsonic region was barely affected. However, the density is still far too low for any reliable predictions to be made about the configuration of the eventual continuum flow.

15.5 The satellite contamination problem

The final application of program DSMC2A deals with the partial return of molecules that are outgassed from a circular disk to the disk as a result of collisions with a supersonic ambient stream. Outgassing is simulated by the superimposition of a zero (macroscopic) velocity jet onto the surface of the disk. For cases that involve realistic outgassing rates from satellites in orbit, the densities in this jet are so low that the 'self-scattered return flux' due to collisions between outgassed molecules is completely absent.

Bird (1981) presented DSMC calculations and an approximate theory for this problem. The theory suggested that the 'return ratio' should follow an equation of the form

$$\frac{N_r}{N_b} \propto \frac{\sigma_{bf}}{\sigma_{ff}} \left(\frac{m_b}{m_f}\right)^{1/2} \frac{s_f}{(Kn)_f} \left(\frac{T_f}{T_b}\right)^{1/3}. \tag{15.8}$$

The subscript $_b$ denotes the outgassed molecules and $_r$ the outgassed molecules that return to the surface, while $_f$ denotes the freestream molecules. The effect of the molecular model enters through the ratio of the cross-section for collisions between a freestream and outgassed molecule and the cross-section for collisions between freestream molecules. The results were generally in accordance with eqn (15.8), although the effect of mass ratio was closer to a ¼ power law.

These previous calculations were severely restricted by the limitations of the computer that was used, and were made for Knudsen numbers in the range 1 to 10. These are relevant to re-entry vehicles, but are orders of magnitude less than the Knudsen numbers that apply under orbital conditions. It is therefore desirable to make similar calculations for representative orbital conditions.

The orbital altitude is taken as 360 km and the atmosphere may be assumed to be atomic oxygen at a number density of 3×10^{14} m^{-3} and a temperature of 1000 K. The outgassing rate at the surface is specified as 5×10^{-10} kg m^{-2}s^{-1} of heavy polymer molecules with a molecular weight of 500 and an assumed diameter of 8×10^{-10} m. For simplicity, the satellite is represented by a disk of diameter 1 m normal to the stream and a surface temperature of 300 K. The reference diameter of atomic oxygen is assumed to be 3×10^{-10} m, with $\omega = 0.75$. This leads to a freestream mean free path of 11,600 m. Eqn (4.24) shows that the outgassing rate is equivalent to molecular effusion from a number density of 2.14×10^{13}. This is less than one tenth the freestream density and, since the flowfield dimensions must be large in comparison with the diameter of the disk, there is extreme difficulty in attaining a sufficient sample of outgassed molecules.

This situation calls for weighting factors to be applied to the species but, as discussed in §10.4, these have not been included in the programs in order to avoid random walk effects. Random walks are not a danger in a near free-molecule flow such as this, and weighting factors can be effectively introduced through the following computational artifice. The freestream number density is reduced by, in this case, a factor of 10,000 while the diameter of the freestream molecules is increased by a factor of 100. The increase in cross-section exactly matches the reduction in density so that an outgassed molecule has the correct probability of collision with a freestream molecule. The molecular mass is unchanged so that the collision mechanics are not affected. However, the velocity components of the freestream molecule should only be changed in one in every 10,000 collisions with an outgassed molecule. This is readily achieved through temporary modifications to the subroutine ELASTIC. Collisions between freestream molecules are realistic because the modifications have no net effect on the freestream collision rate. With these changes, there were approximately equal numbers of outgassed and freestream molecules in the flowfield and the calculation became easy. The flowfield extended to a radius of 6.25 m and to 12.5 m upstream of the disk with a 50×100 cell grid.

It was found that 565 out of 3,282,508 outgassed molecules returned to the surface. The return ratio is therefore 1 in 5,800 and, when substituted into eqn (15.8) this leads to a constant of proportionality $c = 0.075$. This is consistent with the $c = 0.092$ that was obtained from the earlier study for a sphere. This indicates that eqn (15.5) might be useful for the satellite as well as the low altitude case. A similar calculation for outgassed molecules with an atomic weight of 50, rather than 500, verified that the mass ratio power law of ½, rather than ¼, is applicable in this case. Additional calculations with higher and lower freestrean densities also verified the inverse linear dependence on the Knudsen number.

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